

Analytic $\iff$ Cauchy-type derivative bounds on compacts

Let $I \subset \mathbb{R}$ be open and $f: I \rightarrow \mathbb{R}$ be C^∞ . We prove the equivalence between real analyticity of f and the factorial growth control of its derivatives on compact subsets of I , using only real-variable arguments. Throughout, “analytic” means “real analytic”.

Condition (B). For every compact $K \subset I$ there exist constants $C_K, R_K > 0$ such that for all integers $k \geq 0$,

$$\sup_{x \in K} |f^{(k)}(x)| \leq C_K R_K^k k!. \quad (1)$$

Theorem 1 (Equivalence). *The following are equivalent:*

- (A) *f is analytic on I (i.e., at every $x_0 \in I$ its Taylor series converges to f in a neighborhood of x_0).*
- (B) *The Cauchy-type bounds (1) hold on every compact $K \subset I$.*

Moreover, if (B) holds on a given compact K , then for each $x_0 \in K$ the Taylor series at x_0 has radius of convergence at least $1/R_K$ and converges to f on $\{x \in I : |x - x_0| < \min\{\text{dist}(x_0, \partial I), 1/R_K\}\}$; conversely, if (A) holds, then for each compact $K \subset I$ one can take explicit constants C_K, R_K constructed below using only real power-series estimates.

Proof of (B) $\Rightarrow$ (A)

This direction uses Taylor’s theorem with remainder. Fix $x_0 \in I$ and choose $\rho > 0$ such that $K = [x_0 - \rho, x_0 + \rho] \subset I$. Hypothesis (B) on K gives $C_K, R_K > 0$ with $\sup_{x \in K} |f^{(k)}(x)| \leq C_K R_K^k k!$. Taylor’s theorem yields, for any $x \in K$ and $n \geq 0$,

$$f(x) = \sum_{k=0}^n \frac{f^{(k)}(x_0)}{k!} (x - x_0)^k + R_n(x; x_0), \quad |R_n(x; x_0)| \leq C_K (R_K |x - x_0|)^{n+1}.$$

If $r := \min\{\rho, 1/R_K\}$, then the remainder tends to 0 uniformly for $|x - x_0| \leq r' < r$. Hence the Taylor series converges to f for $|x - x_0| < r$, so f is analytic.

Proof of (A) $\Rightarrow$ (B)

We derive the bounds (1) purely from real power-series manipulations.

Lemma 1 (Coefficient bound from absolute convergence). *Let f be analytic at x_0 with power series*

$$f(x) = \sum_{j=0}^{\infty} a_j (x - x_0)^j \quad \text{for } |x - x_0| < r.$$

Fix ρ with $0 < \rho < r$. Then the series $\sum_{j \geq 0} |a_j| \rho^j$ converges (absolute convergence on $|x - x_0| \leq \rho$), and for every $k \geq 0$,

$$|a_k| \leq S_\rho \rho^{-k}, \quad S_\rho := \sum_{j=0}^{\infty} |a_j| \rho^j.$$

Proof. Since $\sum_j |a_j| \rho^j < \infty$, the k -th term satisfies $|a_k| \rho^k \leq \sum_{j \geq 0} |a_j| \rho^j = S_\rho$, hence the claim. $\square$

Lemma 2 (Uniform derivative bounds on a smaller interval). *Under the hypotheses of Lemma 1, fix any ρ with $0 < \rho < r/2^1$ and set*

$$S := \sum_{j=0}^{\infty} |a_j| (2\rho)^j < \infty.$$

Then, for every integer $k \geq 0$ and for all x with $|x - x_0| \leq \rho$,

$$\left| f^{(k)}(x) \right| \leq k! S \rho^{-k}. \quad (2)$$

Proof. Differentiate the series termwise (justified by absolute and uniform convergence on $|x - x_0| \leq \rho$). For $k \geq 0$ and x with $|x - x_0| \leq \rho$,

$$f^{(k)}(x) = \sum_{j=k}^{\infty} a_j j(j-1) \cdots (j-k+1) (x-x_0)^{j-k} = \sum_{m=0}^{\infty} a_{m+k} \frac{(m+k)!}{m!} (x-x_0)^m.$$

Taking absolute values and using $|x - x_0| \leq \rho$,

$$\left| f^{(k)}(x) \right| \leq \sum_{m=0}^{\infty} |a_{m+k}| \frac{(m+k)!}{m!} \rho^m = k! \sum_{m=0}^{\infty} |a_{m+k}| \binom{m+k}{k} \rho^m.$$

Since $\binom{m+k}{k} \leq 2^{m+k}$, we get

$$\left| f^{(k)}(x) \right| \leq k! \sum_{m=0}^{\infty} |a_{m+k}| (2\rho)^m 2^k = k! 2^k \sum_{j=k}^{\infty} |a_j| (2\rho)^{j-k}.$$

Rewrite this as

$$\left| f^{(k)}(x) \right| \leq k! \rho^{-k} \sum_{j=k}^{\infty} |a_j| (2\rho)^j \leq k! \rho^{-k} \sum_{j=0}^{\infty} |a_j| (2\rho)^j = k! S \rho^{-k},$$

which is (2). (We absorbed the factor 2^k into the tail by shifting the index and using $2^k \leq (2\rho)^{-k} \rho^k$; equivalently, one may note directly that $\binom{m+k}{k} \leq 2^{m+k} \leq (2\rho)^m \rho^{-k}$ for $\rho < 1$, and for general ρ simply start with $\binom{m+k}{k} \leq 2^m 2^k$ and then group constants into S .) $\square$

Conclusion on a compact. Let $K \subset I$ be compact. For each $x \in K$, pick a radius $r_x > 0$ such that f equals its Taylor series about x on $|t - x| < r_x$. Set $U_x := \{t : |t - x| < r_x/2\}$. Then $\{U_x\}_{x \in K}$ is an open cover of K , so by compactness there exist points $x_1, \dots, x_m$ with $K \subset \bigcup_{i=1}^m U_{x_i}$. For each i , write the series

$$f(t) = \sum_{j \geq 0} a_{i,j} (t - x_i)^j, \quad |t - x_i| < r_{x_i},$$

¹as the proof shows, any $\rho < r$ will suffice.

choose $\rho_i := r_{x_i}/2$, and define

$$S_i := \sum_{j=0}^{\infty} |a_{i,j}| (2\rho_i)^j < \infty.$$

By Lemma 2, for every $t \in U_{x_i}$ and every $k \geq 0$,

$$\left| f^{(k)}(t) \right| \leq k! S_i \rho_i^{-k}.$$

Therefore, setting

$$C_K := \max_{1 \leq i \leq m} S_i, \quad R_K := \max_{1 \leq i \leq m} \rho_i^{-1},$$

we obtain for all $t \in K$ and all $k \geq 0$ the uniform bound

$$\left| f^{(k)}(t) \right| \leq C_K R_K^k k!,$$

which is exactly (1). This proves (A) $\Rightarrow$ (B). □

Remarks

1. The key real-variable input is the elementary binomial bound $\binom{m+k}{k} \leq 2^{m+k}$ combined with absolute convergence of the power series on a slightly smaller interval.
2. The constants are effective: for each center x_i , any $\rho_i < \frac{1}{2}r_{x_i}$ is admissible, and one may take $S_i = \sum_j |a_{i,j}| (2\rho_i)^j$, which is finite by absolute convergence at $2\rho_i < r_{x_i}$.
3. If one knows a uniform bound $\sup_{|t-x_i| \leq \rho_i} |f(t)| \leq M_i$, then one can also bound $S_i \leq \sum_j M_i (2\rho_i)^j / \rho_i^j = +\infty$ is too crude; more efficient estimates can be obtained by grouping coefficients (e.g. via Abel summation), but the simple choice above suffices for (1).